Ogbonna Arinze

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SUMMARY

Senior Full Stack Engineer with 8 years of professional experience building high-impact web applications across healthcare, fintech, civic tech, and enterprise environments. Deep expertise in React.js, TypeScript, and modern state management (Redux, Zustand), with proven ability to integrate cutting-edge AI models into production workflows including RAG systems and multi-agent LLM pipelines. Experienced in healthcare technology at Genentech (Roche), building HIPAA-aware clinical trial tools with strong data privacy practices. Skilled in real-time systems (WebSockets, SSE, EventSource), comprehensive test suite development (Playwright, Jest), and end-to-end feature ownership from conception through deployment.

CORE COMPETENCIES

- **Frontend & Desktop:** React.js (hooks, lifecycle, performance optimization), TypeScript, Next.js, Vue.js, Redux, Zustand, Context API, Browser APIs, and cross-platform application development (Chrome extensions).
- **AI & LLM Integration:** RAG pipelines, OpenAI API, LangChain, Vercel AI SDK, Mastra, vector search (pgvector, Chroma), embeddings, multi-agent workflows, and real-time AI-powered features.
- **Backend & Real-time:** NestJS, NodeJS REST APIs, PostgreSQL, MongoDB, Prisma, WebSockets, SSE streaming, EventSource API, JWT authentication, and RBAC.
- **Performance & Quality:** Lazy loading, code splitting, caching strategies, Lighthouse, Chrome DevTools, Playwright, Jest, Cypress, Allure Reports, and CI-driven QA.
- **Infrastructure & DevOps:** Docker, GitHub Actions, CI/CD pipelines, AWS (CloudFront, S3), Nx monorepo orchestration, and automated deployment workflows.
- **Leadership:** Technical architecture decisions, mentorship, Agile/Scrum, design systems, code quality standards, and product-minded development.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)

Frontend Engineer *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension**, building cross-platform desktop integration for clinical site workflows and enhancing adoption across EHR systems.
- Co-developed a full-stack **Next.js chat widget** powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources and query their data in real time via a conversational interface.
- Optimized application performance by implementing **lazy loading, code splitting**, dynamic imports, memoization to minimize re-renders, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and **Chrome DevTools**.
- Implemented **JWT-based authentication** and **role-based access control** with **TypeScript** and **React Query**, using HTTP-only cookies for secure token handling and permission-based UI rendering in a healthcare-compliant environment.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components with hooks and streamlining data flow across the application.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Maintained and extended the existing **Vue.js** codebase during the migration period, fixing bugs and shipping features in parallel with the React rebuild.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality.

Descasio Lagos, Nigeria (Hybrid)

Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing **caching mechanisms** across the React application, significantly boosting speed and user experience.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports and more reliable releases.

- Utilized **Storybook** for documenting React components, streamlining change management and supporting agile development practices with precise user story planning.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3, handling thousands to millions of daily transactions and boosting settlement rates by **80%**.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React** with **Redux** for state management, **RxJS** for data streaming, and **Axios** for asynchronous API requests.
- Implemented **real-time data streaming** using the native browser **EventSource API**, replacing conventional socket.io methods for faster updates and a more responsive interface.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Built dynamic data visualization dashboards using **React** and **react-chartjs** with **Redux Toolkit** and **Ant Design** UI.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, focusing on testable, maintainable, and well-documented code.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** and **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and task management.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app with 22 interactive chart types (Recharts), Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service.
- Designed and implemented a **multi-agent AI system** using **Mastra** with 5 specialized agents and a three-stage intent classification pipeline (keyword scoring, LLM classification with 5-min cache, inferred fallback), powered by a **hybrid RAG pipeline** combining **pgvector** similarity search with BM25 full-text search, Reciprocal Rank Fusion, and **Cohere** reranking with Voyage AI embeddings (708K+ chunks).
- Implemented **SSE streaming** for real-time chat responses with 15s keepalive heartbeats to prevent reverse proxy timeouts, token-budgeted context management, conversation summarization, and memory extraction.
- Built the document ingestion pipeline supporting PDF (with OCR via Tesseract), XLSX/XLS, DOCX, JSON, and Markdown extractors, processing 700+ documents (959 files) across 37 states with zero ingestion errors.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** E2E tests across 4 projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits (API 1GB, Ingest 4GB, Postgres 2GB), and 30s graceful shutdown.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*